

Tingwo Security Company:

Relational Model

The relational model is ideal because it uses a structured format of tables, rows, and columns connected by keys. This provides:

- **Data Integrity: ACID compliance** ensures sensitive inventory and issuance transactions remain accurate and consistent.
- **Complex Relationships:** It is specifically designed to handle the intricate links between guards, supervisors, zones, and stock items.
- **Efficiency: Normalization** prevents data redundancy (e.g., storing supervisor details only once), while **SQL** allows for complex queries and reports.

The company might insist on a non-relational database if their priorities shift toward:

- **Scalability:** These models support **horizontal scaling**, which is necessary if the company expands rapidly to hundreds of thousands of employees across various regions.
- **Flexibility:** A **flexible schema** allows them to easily store unstructured or semi-structured data, such as GPS logs from motorbikes or digital delivery receipts, without predefined tables.
- **Speed: Key-Value or Document models** can offer faster performance for simple lookups, such as a clerk quickly checking if a specific item is currently in stock.